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Masterarbeit

Master's Thesis

presented by / von

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1 Introduction

1.1 REXUS and stuff

1.2 State of the Art

1.3 Objectives

Primary objectives of the MEEGA experiment:

- fluid-mechanical recreation of a plume qualitatively similar to plumes found on enceladus
- characterisation of the created plume

1.4 Outline

2 Subproblem: Liquid Infiltration

The first subproblem deals with liquid infiltrating the evaporation chamber. The injection system assembly in the experiment consists of:

- the hydraulic accumulator, consisting of two volumes separated by a diaphragm and occupied by low-pressure nitrogen and water respectively.
- a pivoted armature valve enabling the flow of water from the accumulator into the cross manifold
- the cross manifold to distribute incoming fluid flow
- the check valves which finally allow the fluid to flow into the evaporation chamber.

During rocket ascent, the internal pressure of the evaporation chamber is reduced down to 0.03 MPa. Shortly before the apogee is reached, the injection assembly starts injecting the fluid into the evaporation chamber. The check valves cut off the flow to the evaporation chamber when the internal pressure rises to 0.05 MPa.

2.1 Objective

Study the flow and the transient temperature field in the evaporation chamber (for itself and to estimate the intial conditions that would be present in the next subproblem)

2.2 Problem Description and Background

In order to validate the continuum hypothesis in the evaporation chamber, the Knudsen number is estimated. The Knudsen number, defined as the ratio between the mean free path and a scale length of a selected macroscopic flow property. Assuming that the air present in the chamber is dry with a molar mass $28.96\,\mathrm{g}\cdot\mathrm{mol}^{-1}$ and behaves as an ideal gas, the mean free path, λ, can be estimated using the dynamic viscosity, density, and temperature of the air in the evaporation chamber. Using the remarks provided in the SED, the temperature at start of the experiment inside the evaporation chamber can be assumed to be 280 K in the worst case. Thus using Sutherland's relation and standard sea-level conditions for dry air, the dynamic viscosity, μ results in $1.716e-5Pa.s$. Consequently, the mean free path,

$$\lambda = \frac{\mu}{\rho}\sqrt{\frac{\pi m}{2k_B T}} \approx 2.0732e-06\,\mathrm{m} \tag{1}$$

The Knudsen number, K, is thus of the order 10^{-5}, confirming the continuum assumption.

- [REYNOLDS NUMBER]
- [ATWOOD NUMBER?]
- [INITIAL CONDITIONS]
- [ESTIMATE INITIAL CONDITIONS: WATER MASS FLOW RATE - how to assume inflow to be comprised of two inflows instead of three]
- [STOPPING CRITERIA]

The simulation assumes a turbulent and unsteady flow. The problem is modelled on an axisymmetric 2D domain as depicted in Figure [FIGGIF].

2.2.1 Computational Methodology

The simulation assumes a turbulent and unsteady flow, utlising the SIMPLE implicit scheme to solve for the solution variables. The computational domain is discretised using an orthogonal quadrilateral mesh. Similar to the validation studies, the convective and diffusive terms are discretised using the second-order upwind scheme.

The least squares method is used to estimate the values of the gradients at the face centres and the VOF solver uses the first-order HRIC scheme to reconstruct the interface. As the simulation emulates the Rayleigh-Taylor instability at high Atwood numbers, the modified K-Omega model is used to model turbulence. The transient term is discretised using the first-order implicit Euler-method.

2.3 Validation Problems

2.3.1 Stefan Problem

Phase changes can be regarded as the growth of the domain of one phase accompanied by the reduction of the domain of the other phase. This growth or reduction in domain sizes can be triggered by thermal as well as non-thermal processes [shumon koga] and the representative mathematical dynamic model containing moving boundaries is known as the Stefan system. The associated problem of analysing and finding the solutions to the aforementioned model is known as the Stefan problem. [koga]. The classical Stefan problem is a one dimensional solidification or melting problem citelaterjonas, jonsson. The one-dimensional Stefan problem under semi-infinite geometry, uniform initial temperature, constant boundary temperature, and thermophysical properties constant in each phase is probably the simplest, explicitly solvable problem of phase change [lectures, uni tenessee knoxville lectures], serving as a good benchmark to validate the phase change VOF solver in Star-CCM+. [DOES IT ALSO SERVE AS A GOOD BENCHMARK FOR SIMPLE HEAT CONDUCTION TOO??? BECOZ TWO BIRDS ONE STONE]

Assuming that the region in space, $\Omega \subset \mathbb{R}^2$, occupied by the material is the 2D semi-infinite rectangle $\Omega = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x \in [0,\infty), y \in [0,\Delta h]\}$, with the solid phase and the liquid phase occupying the regions Ω_s and Ω_l respectively such that $\Omega = \Omega_s \cup \Omega_l$. At any given time, t, the two phases are separated by the interface, $\Gamma(t)$, that marks the change in phase of the material. The boundary of the domain Ω is denoted by $\partial\Omega$. The problem space, $\mathcal{D}$, is the spatial and temporal domain occupied by the material, $\mathcal{D} = \{[t_0, t_f] \times \Omega\}$. With the heat equation governing the temperature field in the liquid region, the Stefan condition at the interface, and the respective boundary conditions, the system of equations defining the one-dimensional Stefan problem can be written as,

$$\begin{cases} \frac{\partial T}{\partial t} - \alpha \frac{\partial^2 T}{\partial x^2} & = 0, \\ \rho h_m \frac{d\Gamma(x(t),t)}{dt} + \kappa \frac{\partial \Gamma(x(t),t)}{\partial x} & = 0, \\ T(x,0) & = T_l, \\ \nabla T(x,t) \cdot \mathbf{n} & = 0, \\ T(x,0) & = T_m \end{cases} \tag{2}$$

- the system of equations is well-posed and has a unique solution. the evolution of the temperature field along with the evolution of the phase change interface is studied. The simulation assumes laminar and unsteady flow and solves for the solution variables implicitly using the SIMPLE scheme. The 2D region is discretised using a uniform orthogonal quadrilateral mesh of varying sizes in order to study the spatial convergence of the solution. The physical parameters for the simulation are selected according to water ice, as in [boledi et al]. A constant density $\rho = 1000\,\mathrm{kg \cdot m^{-3}}$, dynamic viscosity $\mu = 8.8871e-4\,\mathrm{Pa \cdot s}$, latent heat of fusion $100000\,\mathrm{J \cdot kg^{-1}}$, and specific heat $4200\,\mathrm{J \cdot kg^{-1}K^{-1}}$. Thermal conductivities are defined as $2.33\,\mathrm{W \cdot m^{-1}K^{-1}}$ and $0.62\,\mathrm{W \cdot m^{-1}K^{-1}}$ for the liquid and solid phases respectively. All four boundaries are assigned a no-slip shear stress specification. The boundary at $x = 0$ is assigned a constant temperature, $T = 273\,\mathrm{K}$, while the other boundaries are assumed to be adiabatic. The convection and diffusion terms are discretised using the second-order upwind scheme and the least squares method was used to interpolate the values of the gradients from the cell centres onto the face centres. The transient term is discretised using the implicit Euler-method which is first-order accurate. First-order HRIC is used to reconstruct the interface. Since Star-CCM+ sets under- and over-relaxation factors conservatively, the values are left unchanged.

The evolution of the temperature field for the simulation with cell size, $\Delta h = 5e-3$ is shown in Figure [FIG]. A clear propagation of the interface from the left to right is observed with increasing simulation times. The temperature also gradually decreases in the liquid phase from the Dirichlet boundary at $x = 0$ towards the melting temperature, T_m, at the interface after which it stays constant.

- 'This is exactly what one would expect for a velocity that is decreasing in time as is the case for the classical Stefan problem. One can also see that all simulation results converge to the analytical solution for a finer mesh and a smaller time step.'
- [ABOUT CONVERGENCE]
- [ABOUT COMPARISON OF TEMPERATURE W ANALYTICAL SOLUTION]
- [ABOUT COMPARISON OF PCI W ANALYTICAL SOLUTION]

2.3.2 Rayleigh-Taylor Instability

When two superposed fluids with differing densities are accelerated in a direction perpendicular to their interface, the surface is unstable when the heavier fluid is accelerated into the lighter fluid. [ref 15, p192 - Birkhoff, 1954]. The instability arising therefrom is known as the Rayleigh-Taylor (RT) instability. The growth of perturbations at the interface is characterised by opposing pressure and density gradients [Banerjee 2020, Schilling et all 2020, HutchinsonDissertation] and the density difference is represented by the *Atwood number*, $At = (\rho_H - \rho_L)/(\rho_H + \rho_L)$. Although not encountered in its ideal form, the RT instability may occur across various natural and technological processes where it is often modified by the presence of additional processes [mention which processes?]. The principles governing the nonlinear RT instability also govern any motion of one fluid in another of different density [Tryggvason rt instability], thus serving as a good measure to study the infiltration of the evaporation chamber in the first subproblem. The instability has moreover served as a classical benchmark for the evaluation of the behaviour of two phase flows [Goulding et al.] ['WHY' imminent].

We assume that the two immiscible, viscuous, and incompressible fluids occupy the regions,

$$\Omega_i(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : (-1)^i (y - h(t,x)) > 0,\ t \geq 0\},\ i = 1,2$$

and are separated by the interface, $\Gamma(t)$ where

$$\Gamma(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : y = h(t,x),\ t \geq 0\}$$

Using the navier stokes equations for incompressible fluids, a mixture density and viscosisty, the jump conditions at the interfaces, and the initial boundary conditions, the governing equation for the Rayleigh-Taylor instability can be written as

$$\begin{cases} \rho \frac{\partial(\mathbf{u})}{\partial t} + \rho \nabla \cdot (\mathbf{u}\mathbf{u}) & = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{g}\rho, \\ \nabla \cdot \mathbf{u} & = 0, \\ [\![S(u, p + \mathbf{g}\rho)\mathbf{n}]\!] & = \sigma\kappa\mathbf{n} + [\![\rho]\!]\mathbf{g}y \\ [\![u]\!] & = 0 \\ V & = \mathbf{u} \cdot \mathbf{n} \\ u(0) & = u_0, \\ \Gamma(0) & = \Gamma_0 \end{cases} \tag{3}$$

where ρ is the density, $\mathbf{u}$ is the velocity vector, p is the pressure, μ is the dynamic viscosity, and g is the gravitational acceleration.

In a system of two fluids with a perturbed interface, the heavier fluid penetrates the lighter fluid due to gravitation thereby pushing the lighter fluid upwards. The initial perturbations, generally studied using linear stability analysis, lead to an exponential growth [andrews and dalziel] and transition into non-lineartiy resulting in the rise

and sink of the bubbles of lighter fluid and spikes of the heavier fluid respectively [sharp, bannerjee]. Figure [name figure] illustrates the parameter definitions and nomenclature of an evolving single-mode [need i define single mode rt?] RT instability interface.

The simulation assumer turbulent and unsteady flow, utilising the SIMPLE implicit scheme. The computational domain is discretised using an orthogonal quadrilateral mesh. The convection and diffusion terms are discretised using the second-order upwind scheme while the least squares method is again used to interpolate the values of the gradients from the cell centres onto the face centres. The VOF solver utilises the first-order HRIC scheme to reconstruct the interface. The transient term is discretised using the first-order accurate implicit Euler-method. The modified K-Omega (SST k-omega) model is used to account for turbulence and surface is treated semi-implicitly. [MENTION INITIAL CONDITIONS] The instability is simulated on a 2D rectangular domain, $[0, L_x] \times [0, 4L_x]$, where L_x is the reference length. The initial interface, located at y_0, represents a single-mode perturbation with amplitude, $A = 0.1L_x$, and wavelength, $\lambda = L_x$.

$$\Gamma_0 = 2L_x + A\cos\left(\frac{2\pi}{L_x}\right)$$

The lower limit of the high Atwood number regime requires the Atwood number to be atleast 0.5 [1]. Gravitational acceleration acts in a downward direction with a magnitude of $g = 1.0$. Periodic boundary conditions are enforced in the horizontal direction and slip wall conditions are applied at the top and bottom boundaries. The physical parameters are selected to make the simulations dynamically similar to the ones studied in [2].

According to [3], a reasonable solution to the RT instability with $At \approx 0.75$ can be obtained with a 128×512 mesh. With similar mesh resolutions being used in [2, 4] to study similar RT instabilities, in addition to a mesh convergence study being performed, the domain is discretised using a uniform orthogonal quadrilateral mesh with size $\delta_h = 0.0045$. The timestep for the simulations is selected as $\delta t = 2.5 \times 10e-4/\sqrt{At}$ similar to [5, 2]. The following cases are studied:

Case	At	Re	σ	δ_h
1	0.5	300	$1e-5$	0.05
2	0.5	300	$1e-5$	0.007
3	0.5	300	$1e-5$	0.0045
4	0.5	300	$1e-5$	0.0025
5	0.2	1000	$1e-5$	0.0045
6	0.5	1000	$1e-5$	0.0045
5	0.5	3000	$1e-5$	0.0045
6	0.5	3000	$1e-5$	0.0045

- [ABOUT SPATIAL AND TEMPORAL CONVERGENCE]
- [ABOUT EVOLUTION OF INTERFACE W VARYING REYNOLDS AND VARYING ATWOOD:] Figure [FIGFIG] and figure [GIFGIF] depict the evolution of the interface at different times for varying density ratios and Reynolds numbers respectively as simulated in Star-CCM+. Upto high Reynolds numbers, the temporal evolution and the structural features of the interface agree qualitatively with the results published in [hamzelhoom huang, ding]. As the solution evolves, the penetration of the spike increases and it eventually rolls up and forms two transient counter-rotating vortices due to shear forces and Kelvin-Helmholtz type instabilities [hamzelhoo], which go on to contribute to turbulent mixing and chaotic flow features [Sharp]. [WHY DO HIGH Re NUM SIMS DEVIATE FROM EXPECTED. MENTION SALIENT FEATURES OF THE EVOLUTION] Simulations conducted for Atwood numbers very close to one also qualita-

tively show a similar progression and evolution of structures as the simulations in [huang]. [What happens at high atwood numbers]

- [ABOUT COMPARISON OF BUBBLE HEIGHTS W DNS DATA AND L2 AND L-INF ERRORS:] Using the bubble and spike heights, defined as the maximum vertical distance of the spike and bubble as measured from the undisturbed initial interface, the results can be quantitatively compared to [hamzelhoo] as shown in Figure **??**. Table **??** summarises the root mean square errors and the maximum errors between the simulated results and [hamzelhoo].

2.3.3 Liquid Infiltration of Evaporation Chamber

- [ABOUT CONVERGENCE]
- [ON EVOLUTION OF INTERFACE and FLOW FIELD and RESULTING HEIGHT OF WATER]
- [ON EVOLUTION OF TEMPERATURE FIELD]

2.4 Conclusion?

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